

ZACHARY A. BOLTON

Full-Stack & Machine Learning Engineer



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ZacharyABolton



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Portfolio

I build products end to end — from years leading technology at an EdTech company to engineering Boostlingo's real-time voice AI in production. I pair product pragmatism with rigorous ML methodology: eval harnesses, competitive benchmarks, and a publicly released accuracy study. Happiest taking ambiguous zero-to-one problems from architecture to deploy.

EXPERIENCE

BOOSTLINGO - August 2025 to Present SENIOR MACHINE LEARNING ENGINEER

Language-services technology company delivering human and AI interpreting to healthcare, legal, and business clients worldwide.

- Lead engineer on Boostlingo's real-time AI Interpreter (.NET/Orleans): built voice-activity detection, crosstalk handling, and conversation-management mechanics for live interpreted calls
- Built the team's LLM evaluation infrastructure from scratch: translation-quality harness with GEMBA scoring, multilingual datasets and baselines, wired into CI
- Co-led **publicly released AI interpreting accuracy study** benchmarking medically complex and legal scenarios against competing systems
- Drove hallucination-reduction and correct-language-output initiatives; owned production deploys, feature-flagged rollouts, IVR/telephony integrations
- Evaluated PII-redaction vendors (Private.AI, CaseGuard, AssemblyAI) for sensitive healthcare call data

Stack: C#/.NET (Microsoft Orleans), Python, LLM & speech APIs (Gemini et al.), Twilio, Datadog LLM Observability, Git/CI-CD

UNIVERSITY OF LONDON - April 2024 to April 2025 INDEPENDENT ML RESEARCHER

- Implemented **Siamese Sentence-BERT authorship-verification architecture** (AUC 0.900, F1 0.785 on VALLA benchmark vs PAN 2020 participants); developed novel bias-controlled LILA dataset achieving AUC/F1/C@1 > 0.99
- Deployed end-to-end data-processing/ML pipeline to AWS EC2 handling 10M+ words

Stack: PyTorch, HuggingFace Transformers, Python Scientific Stack, AWS EC2/S3

KID CHESS, INC. - January 2015 to August 2025 SOFTWARE ENGINEER AND DIRECTOR OF TECHNOLOGY

- Built and led **full-stack online learning platform** from its 2020 launch (Angular, TypeScript, Azure), serving ~4,000 students; created GPT-4-powered personalized feedback system
- Automated business processes with Python pipelines (40% efficiency gain); led pandemic transition to fully online services, achieving 90% student retention

Stack: Angular, TypeScript, Azure, Python, OpenAI API, SQL

EDUCATION

- **University of London: First Class Honors** BSc in Computer Science (Machine Learning & AI) | GPA: 90% | April 2025

ML/AI Skills:

- Real-time Voice AI
- LLM Evals & Benchmarking
- PyTorch & HuggingFace
- NLP & Speech Processing
- LLM Observability
- Experiment Design

Programming:

- Python
- TypeScript/JavaScript
- C#/.NET (Orleans)
- Angular
- SQL
- Azure & AWS

Soft Skills:

- 0→1 Product Ownership
- Technical Writing
- Cross-team Collaboration
- Project Leadership
- Remote Collaboration